

State Is the Interface: Evaluating Handoff Fidelity in Multi-Agent LLM Workflows

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Abstract

Multi-agent language-model systems increasingly transfer task ownership between specialized agents, yet existing evaluations rarely measure which task-critical state survives the transfer. We introduce HandoffBench, a controlled evaluation protocol for inter-role handoffs in interactive, policy-constrained tool workflows. Each episode fixes an upstream trace and annotates a hidden atomic state at the handoff boundary, including verified evidence, epistemic unknowns, scoped consent, and policy decisions. Deterministic probes enable separate measurement of source extraction, transport, and receiver utilization errors and link field-level fidelity to downstream regression and policy violations. We compare full history, free-form summaries, structured payloads, and Executable Handoff Capsules, which bind typed claims to trace provenance and machine-checkable preconditions. We pre-register a factorial evaluation to isolate the effects of typing, provenance, and executable checks under matched evidence and budgets. This framework treats state as the interface between agents and makes handoff failures independently measurable rather than conflating them with routing, memory, or general reasoning failures.

1 Introduction

Handoffs are becoming a standard orchestration primitive: an intake agent may collect constraints, a policy agent may verify eligibility, and a resolution agent may execute the final action. The routing decision can be correct while the workflow still fails. A verified fare can disappear, an unresolved field can become an assumed value, or approval for one action can be mistaken for consent to another. These are failures of the interface between agents.

Existing agent benchmarks richly measure tool use, policy compliance, and team performance (Liu et al. 2024; Yao et al. 2024; Lu et al. 2025; Zhu et al. 2025), while recent coding-agent takeover evaluation measures the effort needed to resume interrupted work (KC and Budathoki 2026). These works do not, however, expose a gold semantic state at an inter-role boundary and score which atomic facts, unknowns, permissions, and preconditions crossed it. Consequently, an

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end-to-end failure cannot be localized to source extraction, the transfer representation, or the receiving agent’s use of correctly transferred state.

We present HandoffBench, a benchmark and intervention protocol that holds the upstream trace, target model, tools, and budget fixed while intervening on the handoff representation. Its central object is a task-critical annotated gold state whose claims are typed and linked to authenticated user, tool, policy, or environment events. Core metrics are computed from typed equality, provenance integrity, simulator transitions, and Boolean success predicates rather than an LLM judge.

Our intended contributions are: (1) a controlled benchmark for field-level handoff fidelity in interactive tool workflows; (2) a three-stage diagnostic decomposition of extraction, transport, and utilization, including oracle interventions that identify handoff-induced regression; and (3) a matched factorial protocol that separates typing, provenance, and advisory executable checks from action enforcement. We report development diagnostics transparently but reserve benchmark-level claims for the frozen confirmatory evaluation.

2 Method

Boundary-state evaluation. HandoffBench isolates a single transfer of control rather than evaluating generic conversational memory. An episode is $e = (E, \mathcal{A}, \mathcal{U}, \tau, b, a_s, a_t, G_b, \Phi)$, where E is a seeded workflow environment, τ is the authenticated source trace cut at boundary b , a_s and a_t are the source and receiver roles, G_b is evaluator-private boundary state, and Φ is a machine-executable terminal predicate. For transfer mechanism m , the receiver observes only $V_m(\tau)$ and produces post-boundary trace ρ_m . Across mechanisms we hold fixed E, τ, b , receiver model and prompt, tools, decoding parameters, and seed. Thus, comparisons vary the transfer representation while preserving the evidence and downstream capability.

The evaluator proposal is a task-critical set of typed atomic claims,

$$G_b = \{g_i = (k_i, v_i, q_i, p_i, \ell_i, w_i)\}_{i=1}^n, \quad (1)$$

with canonical key k_i , normalized value v_i , epistemic status $q_i \in \{\text{KNOWN}, \text{UNKNOWN}, \text{CONTRADICTED}, \text{N/A}\}$, authenticated provenance p_i , criticality ℓ_i , and preregistered weight

$w_i > 0$. A proposed claim is retained only when its omission or mutation can change a legal next action or Φ ; open status is itself state and is never imputed. Executable preconditions remain in action rules, and tool-execution metadata is scored through workflow events, so neither is counted again in the primary semantic-state score. Candidate families must be independently double-annotated and adjudicated before sealing; model outputs are unavailable during this process.

Judge-free evaluator. Every receiver first emits the same schema-constrained state probe and then one schema-constrained action. Values are compared by declared canonicalizers (e.g., ISO dates, normalized identifiers, and unordered sets), never by an LLM judge. Within each (category, k) group, gold and predicted claims are assigned one-to-one by deterministic maximum-weight bipartite matching. Hence a duplicated prediction can receive credit at most once and all remaining copies are false positives; duplicated gold atoms likewise require distinct predictions. For a matched pair $(g_i, \hat{g}_j)$, let r_{ij} be overlap measured relative to the gold value and p_{ij} overlap measured relative to the prediction; both reduce to exact equality of status and normalized value for non-set claims, while set claims use directional element overlap. We report the resulting weighted micro diagnostics,

$$R_s = \frac{\sum_{(i,j) \in \mathcal{M}} w_i r_{ij}}{\sum_i w_i}, \quad P_s = \frac{\sum_{(i,j) \in \mathcal{M}} \bar{w}_{c(j)} p_{ij}}{\sum_j \bar{w}_{c(j)}}, \quad (2)$$

where $\bar{w}_c$ is a fixed category weight rather than a model-supplied weight. The assignment optimizes the combined directional recall and precision credit. Distinct from these weighted micro diagnostics, headline transfer fidelity is the unweighted mean of per-field F_1 over populated primary semantic fields, using the same field-local one-to-one matching. We additionally report category recall, contradiction among predictions sharing a gold key, unsupported-key rate, and provenance validity.

The simulator checks action arguments, ordering, call limits, public preconditions, and role allow-lists before applying deterministic user or tool transitions. Strict success is $S(e, m) = \mathbf{1}[\Phi(E_{\text{terminal}}, \rho_m)]$. Episode-level safety and efficiency errors include action without scoped consent, unsupported commitment reports, missed preconditions, duplicate stable tool calls, unnecessary re-asks, and role violations. A gold-state oracle uses the same receiver and environment but exposes canonical G_b . We attribute a paired handoff-induced regression only when the oracle succeeds:

$$\text{HIR}(e, m) = \mathbf{1}\{S(e, o) = 1 \wedge S(e, m) = 0\}. \quad (3)$$

This definition avoids labeling failures that persist under perfect transfer as handoff-caused.

Representation Factorial

The mechanism-identification experiment is a representation-only $2 \times 2 \times 2$ factorial: typing (fixed-envelope free-form versus typed atoms) $\times$ provenance (absent versus trace-linked) $\times$ checks (absent versus executable). All eight cells expose the same ten semantic fields.

For each $(e, \text{model}, \text{seed})$ block, the source receives the identical frozen trace, atomic evidence, public predicates, prompt, schema, and token budget. Crucially, the source is called once per block: its immutable generated artifact, raw output, prompt/schema hashes, and logical usage record are reused across all eight cells. A fairness audit rejects an incomplete cube or any block with unequal source output, prompt, schema, or usage hashes.

The factors alter serialization, not source information. Typing renders the same atoms either as $\{k, q, v\}$ records or as deterministic prose within the fixed field envelope. Provenance-off cells omit bindings but do not withhold evidence from the source. Checks-off cells omit evaluated statuses but do not withhold public predicates. In checks-on cells, the framework evaluates those predicates against the source state; it does not introduce evaluator gold. Every raw artifact is retained unchanged. Validation produces a sidecar and, where relevant, visible error annotations; it neither supplies corrected values nor quarantines a claim in the primary experiment.

All factorial cells use advisory execution. Executable checks therefore mean “represent a computed status,” not “block an action.” Action enforcement is an orthogonal binary intervention: the secondary enforced arm holds the all-on representation fixed and reports prevented violations, false blocks, and success. It is not pooled into factorial effects or described as a representation gain. Legacy Full History, Free Summary, Structured Payload, all-on Capsule, and Gold Oracle remain descriptive controls outside the factorial.

Confirmatory Statistics

The preregistered confirmatory population contains 200 independently authored families (40 in each of five domains), conditional on completed annotation, adjudication, overlap audit, and sealing before any confirmatory call. At least two model families are evaluated; model and seed rows are repeated measures nested within task family. Analysis is intent-to-treat (ITT): every scheduled task-model-seed-condition cell remains in its assigned condition. Provider, schema, or parse failures receive zero strict success and state credit, remain in all denominators, and retain their raw calls and costs.

For the balanced factorial, factors are effect-coded $x_f \in \{-1, +1\}$. We estimate all main effects, all two-way interactions, and the three-way interaction; for term T , the balanced contrast is $2 \mathbb{E}[Y \prod_{f \in T} x_f]$. Strict success is primary; state F_1 , critical error, HIR, tokens, calls, and validator cost are secondary. Confidence intervals use percentile bootstrap resampling of independent task families, carrying along all model, seed, and condition rows. Binary paired contrasts use the same family-cluster bootstrap; exact McNemar tests are sensitivity analyses because repeated model/seed observations are clustered. Optional post-confirmatory mixed diagnostic models may include family and model intercepts, but they are not part of the confirmatory evidence or multiplicity family and will be reported only with convergence diagnostics and any fallbacks.

The confirmatory multiplicity family contains (i) Structured versus Gold Oracle strict success and (ii) the advisory-check factorial main effect on strict success. Their two-sided

p -values are Holm-adjusted; the directional checks hypothesis is supported only if its adjusted two-sided interval excludes zero in the positive direction. Typing, provenance, interactions, legacy contrasts, and domain/model subgroups are two-sided secondary or exploratory analyses and cannot be promoted after inspecting test outcomes. Development-only quantities, if shown, are labeled exploratory and reported separately from confirmatory estimates.

3 Benchmark Construction

Unit of construction. HandoffBench targets state transfer at a fixed control boundary rather than routing quality. Each episode fixes an authenticated upstream trace, source and receiver roles, the boundary cut, a public action contract, deterministic mock tools and user replies, and a machine-executable terminal predicate. The evaluator-side boundary state is a set of typed atomic claims with exact trace provenance and epistemic status (known, unknown, contradicted, or not applicable). We include a claim only when changing or removing it can alter a legal next action or terminal success. Goals, user constraints, verified business facts, ordinary unresolved slots, scoped consent, and organizational policy checks are mutually exclusive roles. Executable action preconditions remain in the public contract, and tool-execution metadata remains in the trace; neither is duplicated as a semantic claim for primary state F1.

Candidate-pool generation. We first authored a workflow blueprint—roles, causal trace, authority boundary, plausible alternatives, legal sequence, and first user-impacting action—then instantiated concrete synthetic dates, monetary values, statuses, scopes, option attributes, and opaque identifiers with deterministic generators. Families are the split unit: entity replacement or paraphrase does not create a new family. The resulting *candidate* pool has 200 unique families, evenly divided among travel, commerce, procurement, enterprise IT, and scheduling (Table 1). Six preregistered stressors vary what must cross the boundary: long distractors, conflicting evidence, irreversible actions, multi-step evidence, missing authority, and user revisions. Of the 200 gold-path proposals, 195 contain a user-impacting action.

Construction checks are deterministic and should not be confused with model results. Schema, provenance, uniqueness, argument-grounding, enum-permutation, and action-sequence mutation tests pass, and the authored sequence executes in 200/200 simulators; an exact-copy oracle likewise reaches 200/200. These numbers establish internal executability only, not task validity or agent competence. When the hidden terminal predicate is withheld, only 137/200 public contracts admit a unique legal terminal sequence; clarification or authority actions can commute in 63. Moreover, only 305/990 proposed claims have a direct syntactic counterfactual witness in a guard key or exact action argument. The remainder may be semantically necessary, but static code cannot establish their minimal causal relevance. Both sets therefore require independent human reconstruction. Simple leakage diagnostics obtain 14/200 from the public catalog alone and 12/200 from predicate-only features. A stronger diagnostic is sometimes mislabeled “name only”: it receives

Table 1: Repository-static statistics for candidate v2. “Oracle” denotes deterministic simulator consistency; human validation is pending.

Statistic	Count / rate
Candidate families (five domains)	200 (40 each)
Primary stressors: LD / CE / IA	38 / 35 / 35
Primary stressors: ME / MA / UR	32 / 30 / 30
Proposed atomic claims	990
User-impacting gold paths	195 / 200
Static sequence / copy-oracle success	200 / 200
Unique legal sequence without predicate	137 / 200
Claims with direct syntactic witness	305 / 990
Catalog-only / predicate-only success	7.0% / 6.0%
Oracle names + enum-first success	69 / 200 (34.5%)
Exact trace / action-graph collisions	0 / 0
Normalized topology overlaps	69 candidates
Lexical pairs at Jaccard $\geq .80$	0

the oracle action-name sequence, then guesses the first value in every public enum. It succeeds on 69/200 (34.5%; Wilson 95% CI [28.3, 41.3]), versus 13.9% expected under uniform enum guessing. All successes occur when every correct enum value is first. This reveals combined oracle-name and interface-ordering leakage, not that action names alone solve 34.5%.

Development-overlap audit. We compared the 200 candidates with 54 development episodes after normalizing dates, numeric and opaque identifiers, and task/trace/boundary IDs. There are no collisions in family ID, entity pool, generator seed, exact trace hash, normalized trace hash, or exact action-graph hash, and no trace-content bigram pair reaches the preregistered Jaccard threshold of .80. However, 69 candidate tasks share a *normalized* action-graph hash with at least one development task (552 candidate–development pair groups). This is distinct from the numerically equal 69 oracle-name/enum-first successes: the former diagnoses topology reuse across splits, whereas the latter diagnoses public-interface ordering leakage within candidates. Normalization may over-collapse common schema structure, but it may also reveal genuine workflow similarity. We therefore do not describe the pool as structurally held out or frozen; these cases require human review and rejection/redesign before sealing.

4 Planned Human Validation

Human validation has not yet been executed, so we report no agreement statistic and call no candidate a test item. The preregistered procedure assigns every label-free packet to two annotators in independently randomized order. Annotators see the authenticated trace, boundary roles, public actions, and deterministic tool semantics, but not source labels, evaluator predicates, model outputs, development results, family blueprints, or one another’s responses. Each first derives the exact legal action sequence, then independently reconstructs the minimal claims, typed values and statuses, criticality, and one or more exact (`trace_id`, `source_type`, `field_path`) pointers. They also flag whether irreversible

arguments are inferable and whether catalog order or action names leak the sequence.

Responses are schema-checked, hashed, and locked before comparison; corrections create a new retained version rather than overwriting the original. We will report one-to-one claim precision/recall/F1, exact typed-value and provenance agreement, category/status/criticality agreement, and exact sequence agreement, clustered by family. Cohen’s κ is used only when both annotators employ at least two labels, with raw agreement and denominators always shown. A third annotator receives disagreements only and may reject an ambiguous task. Tasks with secret arguments, non-unique legal sequences, unresolved safety claims, interface-ordering leakage that trivializes the decision, or problematic normalized topology overlap must be rejected or regenerated before any confirmatory model call. Only after adjudication, overlap remediation, final hashes, and a non-overwritable manifest may a later version be called frozen.

5 Closest Related Work

The broad idea of evaluating agent handoffs is not new. Most directly, *Handoff Debt* interrupts coding agents at deterministic repository checkpoints and compares repository-only, raw-trace, summary-note, and structured-note views, measuring task resolution and successor rediscovery cost (KC and Budathoki 2026). We therefore do not claim the first handoff benchmark, controlled takeover, or structured transfer artifact. Our estimand is instead the fidelity of a semantic interface at an inter-role boundary: each episode exposes an evaluator-only set of atomic claims with epistemic status and trace evidence, and tests which claim errors precede policy-invalid or regressive actions. This also differs from state-probe and runtime-guard work (Jia 2026; She, Liang, and Kang 2026) (Table 2). Concurrent work makes the collision sharper. PACT treats inter-agent messages as compact public action-state updates and evaluates accuracy–cost trade-offs (Huang, Wu, and Zhang 2026); AgentAsk annotates edge-level data gaps, referential drift, signal corruption, and capability gaps, then learns when to clarify (Lin et al. 2026); and EntCollabBench objectively scores role-specialized, permission-isolated enterprise workflows in which context transfer is one bottleneck (Yu et al. 2026). HandoffBench does not claim these messaging, repair, or enterprise settings. It contributes evaluator-visible atomic boundary state and matched interventions that localize extraction, transport, and receiver-utilization failures.

6 Development Findings and Design Adaptation

We used six counterfactual travel families only to debug the protocol and set the confirmatory design; these data are excluded from all confirmatory tables. Across Qwen2.5-14B and Ministral-3-14B (48 scheduled episodes per legacy view), strict success was 68.8% for the all-on executable handoff capsule (EHC), 75.0% for the schema-matched Structured payload, 47.9% for Free Summary, 91.7% for Full History, and 93.8% for the Gold Oracle. The EHC–Structured difference was -6.25 percentage points (family-

Work	Primary object	Distinction of HandoffBench
Handoff Debt	Coding takeover and rediscovery under four context views	Gold field-level boundary state; epistemic and authority-bearing claims; oracle-conditional transfer regression rather than repository resumability.
NeuroState-Bench	Commitment integrity through side-query probes within an agent profile	A manipulated predecessor–successor boundary and alternative transport representations, with claims bound to authenticated trace events.
Provenance-Guard	Runtime screening of proposed tool actions using evidence for user intent	Provenance is an advisory representation factor in the primary study; action blocking is isolated as a separate enforcement intervention.

Table 2: Claim-level positioning against the closest collisions. Structured handoffs, state probes, and provenance-aware blocking are prior art; our target is controlled measurement of semantic state transfer across a role handoff.

Development view	Success	Rate
Full History	44/48	91.7%
Free Summary	23/48	47.9%
Structured Payload	36/48	75.0%
All-on EHC	33/48	68.8%
Gold Oracle	45/48	93.8%

Table 3: Development-only strict success across two model families and six independent families (four counterfactual variants each). These are design diagnostics, not benchmark test results.

clustered 95% CI $[-22.9, 12.5]$). This is a negative development result: it does not support generic EHC superiority, and we retain rather than tune it away.

A representation-matched $2 \times 2 \times 2$ development factorial (typing, trace-linked provenance, and advisory checks) scheduled 384 cells for two model families; 381 completed provider/parsing successfully, and all cells remain in the intent-to-treat denominator. Strict-success main effects were $+2.60$ pp for typing (95% CI $[0.52, 4.69]$), -6.77 pp for provenance ($[-15.63, 2.60]$), and $+6.77$ pp for advisory checks ($[3.65, 9.38]$). The provenance–checks interaction was -4.69 pp ($[-10.42, -0.52]$); provenance also increased mean critical-error count by 0.042 ($[0.010, 0.078]$). With only six independent families, these estimates are diagnostic, not population evidence. They changed the study’s thesis from “all-on capsules are superior” to two falsifiable benchmark questions: whether boundary regression exists, and which representation mechanisms help or harm under matched information.

Power was likewise treated as a pre-test design input. A

Factor / interaction	Effect	Family 95% CI
Typing	+2.60 pp	[0.52, 4.69]
Provenance	−6.77 pp	[−15.63, 2.60]
Advisory checks	+6.77 pp	[3.65, 9.38]
Provenance × checks	−4.69 pp	[−10.42, −0.52]

Table 4: Development-only strict-success effects from the fair $2 \times 2 \times 2$ factorial (two model families, six independent families, 384 ITT cells). The source-identity audit passed all 48 episode–model blocks.

fixed-seed simulation with model repeats nested within family and family-cluster bootstrap inference estimated, at ICC = .10, power of .441 for an 8-pp effect and .633 for a 10-pp effect at 120 families; at 200 families the corresponding estimates were .691 and .847. We therefore expanded the candidate pool to 200 independently authored families (40 per domain) before annotation or any candidate model call. These simulated values are assumptions, not observed test effects, and do not permit optional stopping.

7 Confirmatory Analysis Plan

The confirmatory unit is the independently authored task family. Model and seed observations are repeated measures nested within family and are never counted as independent samples. Every scheduled task–model–seed–condition cell is analyzed intent-to-treat: provider, schema, and parse failures receive zero strict-success and state credit while their calls and costs remain in the release artifacts. Strict end-to-end success is primary; first-probe macro state F1, episode-level critical errors, oracle-conditional handoff-induced regression, input/output tokens, calls, and validator cost are secondary.

Two contrasts form the confirmatory family: Structured versus Gold Oracle on strict success, testing whether non-oracle transfer induces boundary regression, and the advisory-check main effect in the matched factorial. Paired binary effects use 95% family-cluster percentile-bootstrap intervals; exact McNemar tests are sensitivity analyses because repeated model/seed rows are clustered. Continuous and count outcomes use the same family bootstrap. The two primary two-sided tests are Holm-adjusted. Typing and provenance effects are two-sided, as development evidence was uncertain or negative; all interactions, domain/category/model slices, and legacy-view contrasts are secondary or exploratory. EHC versus Structured remains a reported two-sided comparison, but no superiority claim is made unless its corrected interval excludes zero in the favorable direction. Advisory representation and enforced action gating are analyzed separately, and null or harmful provenance effects remain reportable outcomes.

8 Ethics and Limitations

All scenarios, entities, traces, policies, and tool outputs are project-authored synthetic data, not customer records or private company policies; no real person is intentionally represented. Nevertheless, future free-text annotations could in-

troduce personal information, so annotators use participant codes, may not paste real credentials or records, and exports require PII/secret review. The tasks simulate consequential purchases, refunds, identity and access checks, accessibility requests, and travel-document workflows. Their simplified policies are unsuitable for operational, legal, medical, financial, employment, or other decisions about real people.

The candidate pool is synthetic and may contain lexical regularities, simplified jurisdictional assumptions, and limited real-world coverage. Static oracle success does not establish human determinacy, and the 34.5% combined oracle-name/enum-first diagnostic and 69 normalized-topology overlaps are material construct-validity threats. Even a validated synthetic split would not establish ecological validity. A post-seal replication can adapt separately licensed Tool-Sandbox scenarios and the telecom subset of τ^2 -Bench (Lu et al. 2025; Barres et al. 2025), but must report each source separately and may not replace rejected primary families after outcomes are observed. Moreover, the repository does not yet declare a top-level data license; no open license should be inferred before release terms are set. Inspecting candidates with models would convert them to development data. Accordingly, this draft reports construction diagnostics only and makes no confirmatory performance, agreement, representativeness, or frozen-benchmark claim.

9 Conclusion

Reliable multi-agent systems require more than correct routing. By treating the handoff boundary as a typed, measurable interface, HandoffBench aims to reveal which state is lost, why downstream behavior regresses, and when evidence-linked and executable transfer artifacts help.

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